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The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d | e | f |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 2.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 3.) | ☐ | ☐ | ☐ | ☐ | | |
| 4.) | ☐ | ☐ | ☐ | ☐ | | |
| 5.) | (essay) | | | | | |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{p, r\}$. Which string of symbols are membered in the set $Form$?

- 1 $\neg(\neg(\neg p \supset r))$
- 0 $(p \equiv \neg r)$
- 0 $\neg(\neg(p \vee r) \wedge r)$
- 1 $\vee$
- 1 $\{p \wedge \neg r\}$
- 0 $\neg\neg r$

2.) Which statements are true for the formula below?

$$(\neg((p \supset \neg q) \vee p) \wedge (p \equiv \neg q))$$

- (a) $(\neg q \equiv p)$ is a direct subformula
- 2 $(p \equiv \neg q)$ is a direct subformula
- 0 $(p \supset \neg q)$ is a subformula
- (d) the formula has only one direct subformula
- 2 there are 12 nodes in the construction tree of the formula
- 1 there are two leaves in the construction tree of the formula

3.) Which interpretations are models of the following set of formulae?

$$\{ (\neg(q \vee q) \supset p), ((p \wedge q) \equiv q) \}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

-1 $\varrho(p) = 0, \quad \varrho(q) = 1$

0 $\varrho(p) = 1, \quad \varrho(q) = 0$

2 $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ (p \supset (q \equiv (q \supset q))), (\neg p \equiv q) \}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

2 $\varrho(p) = 0, \quad \varrho(q) = 1$

(c) $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) When can we say that a set of formulae is unsatisfiable?

Score: 1 / 2.